5(a)	charge / potential (difference)	**M1**
	charge is charge on one plate, <u>and</u> potential is p.d. across the plates	**A1**
5(b)	p.d. across both capacitors = E	**B1**
	$Q_T = Q_1 + Q_2$	**B1**
	$C_TE = C_1E + C_2E$ hence $C_T = C_1 + C_2$	**B1**
5(c)(i)	$[(1 / 22) + (1 / 47)]^{-1} = 15\ \mu F$	**A1**
5(c)(ii)	energy = $\frac{1}{2}CV^2$	**C1**
	$= \frac{1}{2} \times 15 \times 10^{-6} \times 12^2$ $= 1.1 \times 10^{-3}$ J	**A1**
5(c)(iii)	initial p.d. (across 22 μF) $= 12 \times (15 / 22)$ $= 8.2$ V **or** final p.d. across both capacitors $= 6.0 \times (22 / 15)$ $= 8.8$ V	**C1**
	$V = V_0 \exp[-t / (2.7 \times 10^6 \times 15 \times 10^{-6})]$	**C1**
	$6.0 = 8.2 \exp[-t / (2.7 \times 10^6 \times 15 \times 10^{-6})]$ **or** $8.8 = 12 \exp[-t / (2.7 \times 10^6 \times 15 \times 10^{-6})]$ $t = 13$ s	**A1**